

Kubernetes (K8s)

Kubernetes is an open source container orchestration engine for automating deployment, scaling, and management of containerized applications. The open source project is hosted by the Cloud Native Computing Foundation (CNCF).

It provides a scalable and resilient framework for automating the deployment, scaling, and management of applications across clusters of servers.

A SMALL HISTORY OF K8S:

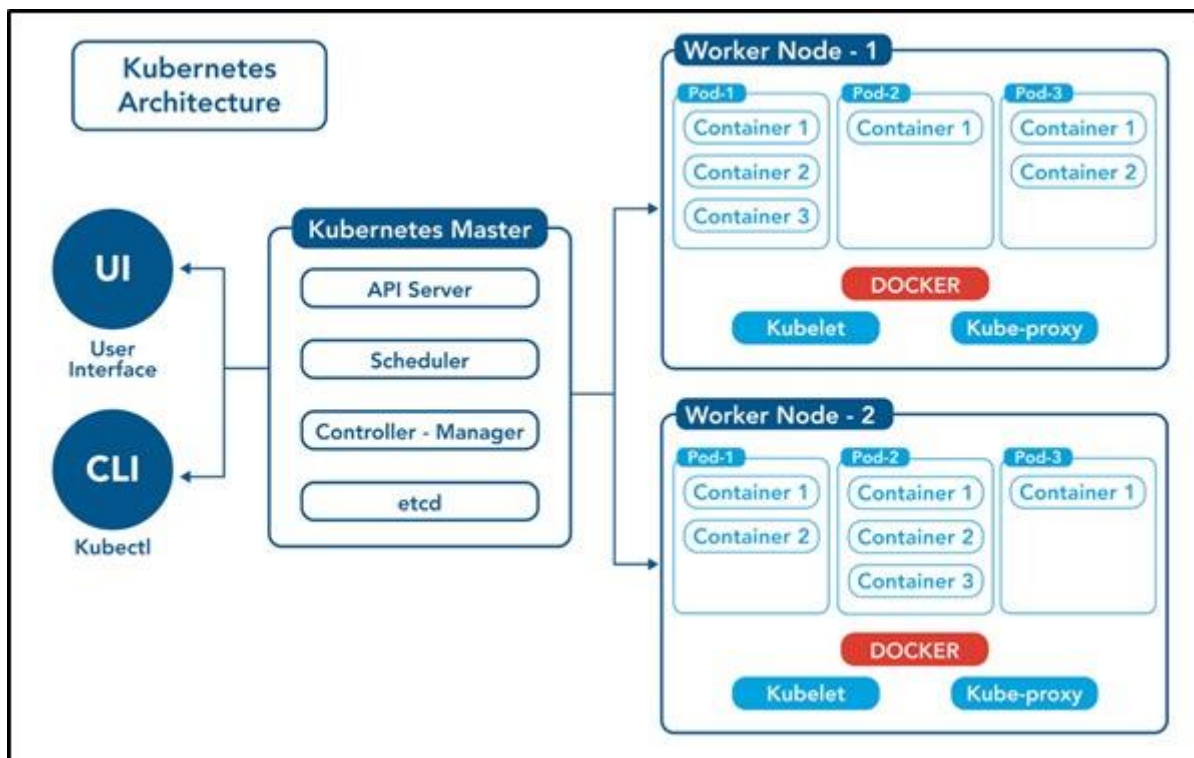
☐ In the early 2000s, Google started developing a system called Borg to manage their internal containerized applications.

☐ Borg enabled Google to run applications at scale, providing features such as automatic scaling, service discovery, and fault tolerance.

☐ In 2014, Google open-sourced a version of Borg called Kubernetes.

☐ Kubernetes was donated to the Cloud Native Computing Foundation (CNCF), a neutral home for open-source cloud-native projects, in July 2015.

☐ Kubernetes 1.8 added significant enhancements for storage, security, and networking. Key features included the stable release of the stateful sets API, expanded support for volume plugins, and improvements in security policies.



Control Plane /Master Node

The control plane's components make global decisions about the cluster (for example, scheduling), as well as detecting and responding to cluster events (for example, starting up a new pod when a deployment's replicas field is unsatisfied).

Control plane components can be run on any machine in the cluster. Do not run user containers on this machine.

Node Components / Worker Nodes

Node components run on every node, maintaining running pods and providing the Kubernetes runtime environment.

1. **Master Node:** The master node is responsible for managing the cluster and coordinating the overall state of the system. It includes the following components:
 - a. **API Server:** The API server is the central control point for all interactions with the cluster. It exposes the Kubernetes API and handles requests from users and other components.

b. Scheduler: The scheduler is responsible for assigning workloads (pods) to individual worker nodes based on resource requirements, constraints, and other policies.

c. Controller Manager: The controller manager runs various controllers that monitor the cluster state and drive it towards the desired state. Examples include the replication controller, node controller, and service controller.

d. etcd: etcd is a distributed key-value store used by Kubernetes to store cluster state and configuration data.

1. Pod: The basic building block of Kubernetes. A pod represents a single instance of a running process within the cluster. It can encapsulate one or more containers that share the same network and storage resources.

BASIC COMMANDS:

1. Create a pod using run command

```
$ kubectl run <pod-name> --image=<image-name> --port=<container-port>
```

```
$ kubectl run my-pod --image=nginx --port=80
```

2. View all the pods

(In default namespace)

```
$ kubectl get pods
```

(In All namespace)

```
$ kubectl get pods -A
```

For a specific namespace

```
$ kubectl get pods -n kube-system
```

For a specific type

```
$ kubectl get pods <pod-name>
```

```
$ kubectl get pods <pod-name> -o wide
```

```
$ kubectl get pods <pod-name> -o yaml
```

```
$ kubectl get pods <pod-name> -o json
```

3. Describe a pod (View Pod details)

```
$ kubectl describe pod <pod-name>
```

```
$ kubectl describe pod my-pod
```

4. View Logs of a pod

```
$ kubectl logs <pod-name>
```

```
$ kubectl logs my-pod
```

5. Execute any command inside Pod (Inside Pod OS)

```
$ kubectl exec <pod-name> -- <command>
```

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: my-app
```

```
spec:
```

```
  containers:
```

```
    - name: my-app-container
```

```
      image: <images>    ports:
```

```
        - containerPort: 9090
```

```
kind: ReplicaSet
```

```
metadata:
```

```
  name: my-rs
```

```
  labels:
```

```
    name: my-rs
```

```
spec:
```

```
replicas: 4
selector:
  matchLabels:
    apptype: web-backend
template:
  metadata:
    labels:
      apptype: web-backend
spec:
  containers:
  - name: my-app
    image:
    ports:
      - containerPort: 8080
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-deploy
  labels:
    name: my-deploy
spec:
  replicas: 4
  selector:
    matchLabels:
      apptype: web-backend
  strategy:
    type: RollingUpdate
```

```
template:
  metadata:
    labels:
      apptype: web-backend
  spec:
    containers:
      - name: my-app
        image:
        ports:
          - containerPort: 7070
```

Deployment.yml

apiVersion: apps/v1

kind: Deployment

metadata:

name: my-deploy

namespace: my-bank

labels:

name: my-deploy

spec:

replicas: 1

selector:

matchLabels:

apptype: web-backend

strategy:

type: RollingUpdate

template:

metadata:

labels:

apptype: web-backend

spec:

containers:

- name: my-web

image: arvindm2004/webapp1:latest

ports:

- containerPort: 9001

apiVersion: v1

kind: Service

metadata:

name: my-service

namespace: my-bank

labels:

app: my-service

spec:

type: NodePort

ports:

- port: 9001

targetPort: 8080

nodePort: 30007

selector:

apptype: web-backend

kubectl create deployment webnginx2 --image=nginx:latest --replicas=1

kubectl scale deploy <deployment-name> --replicas=<desired-replica-count>

Services (short name = svc):

Service is an abstraction that defines a logical set of pods and a policy to access them. Services enable network connectivity and load balancing to the pods that are part of the service, allowing other components within or outside the cluster to interact with the application.

Service Types: Kubernetes supports different types of services:

1. NodePort: Exposes the service on a static port on each selected node's IP. This type makes the service accessible from outside the cluster by the <NodeIP>:<NodePort> combination.
2. ClusterIP: Exposes the service on a cluster-internal IP. This type makes the service only reachable within the cluster.
3. LoadBalancer: Creates an external load balancer in cloud environments, which routes traffic to the service.

2. Create Deployment by executing above YAML file

```
$ kubectl create -f web-deploy.yml
```

Do necessary modifications if exist, else create new

```
$ kubectl create -f web-deploy.yml
```

Completely Modify Pod Template

```
$ kubectl replace -f web-deploy.yml
```

3. View Deployments

```
$ kubectl get deployments
```

```
$ kubectl get deploy
```

```
$ kubectl get deploy -o wide
```

```
$ kubectl get deploy <deployment-name> -o json
```

```
$ kubectl get deploy <deployment-name> -o yaml
```


4. View Deployment Description

```
$ kubectl describe deploy <deployment-name>
```

5. We can modify generated/updated YAML file

```
$ kubectl edit deploy <deployment-name>
```

```
## change replicas: count to any other value then (ESC):wq
```

```
# We can modify our YAML file and then execute apply command
```

```
$ kubectl apply -f web-deploy.yml
```

```
## We can Even scale using command also
```

```
$ kubectl scale deploy <deployment-name> --replicas=<desired-replica-count>
```

6. Delete Deployment

```
$ kubectl delete deploy <deployment-name>
```

```
$ kubectl delete -f web-deploy.yml
```

2. Create ReplicaSet by executing above YAML file

```
$ kubectl create -f rs-test.yml
```

```
# Do necessary modifications if exist, else create new
```

```
$ kubectl apply -f rs-test.yml
```

```
# Completely Modify Pod Template
```

```
$ kubectl replace -f rs-test.yml
```

3. View ReplicaSets

```
$ kubectl get replicaset
```

```
$ kubectl get rs
```

```
$ kubectl get rs -o wide
```

```
$ kubectl get rs <replica-set-name> -o json
```

```
$ kubectl get rs <replica-set-name> -o yaml
```

4. View ReplicaSet Description

```
$ kubectl describe rs <replica-set-name>
```

5. We can modify generated/updated YAML file

```
$ kubectl edit rs <replica-set-name>
```

```
## change replicas: count to any other value then (ESC):wq
```

```
# We can modify our YAML file and then execute apply command
```

```
$ kubectl apply -f rs-test.yml
```

```
## We can Even scale using command also
```

```
$ kubectl scale replicaset <replicaset-name> --replicas=<desired-replica-count>
```

6. Delete ReplicaSet

```
$ kubectl delete rs <replica-set-name>
```

```
$ kubectl delete -f rs-test.yml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: my-deploy
```

```
  labels:
```

```
    name: my-deploy
```

```
spec:
```

```
  replicas: 1
```

```
  selector:
```

```
    matchLabels:
```

```
      apptype: web-backend
```

```
strategy:
  type: RollingUpdate
template:
  metadata:
    labels:
      apptype: web-backend
  spec:
    containers:
      - name: my-app
        image:
        ports:
          - containerPort: 7070
```

```
apiVersion: v1
kind: Service
metadata:
  name: my-service
  labels:
    app: my-service
    type: backend-app
spec:
  type: NodePort
  ports:
    - targetPort: 7070
      port: 7070
      nodePort: 30002
  selector:
```

apptype: web-backend

apiVersion: apps/v1

kind: Deployment

metadata:

name: my-deploy

labels:

name: my-deploy

spec:

replicas: 1

selector:

matchLabels:

apptype: web-backend

strategy:

type: RollingUpdate

template:

metadata:

labels:

apptype: web-backend

spec:

containers:

- name: my-app

image:

ports:

- containerPort: 9000

apiVersion: v1

kind: Service

metadata:

name: my-service

labels:

app: my-service

spec:

type: NodePort

ports:

- port: 9000

targetPort: 8080

nodePort: 30002

selector:

apptype: web-backend

Namespace (short name = ns):

namespace is a virtual cluster or logical partition within a cluster that provides a way to organize and isolate resources. It allows multiple teams or projects to share the same physical cluster while maintaining resource separation and access control.

To create a namespace:

\$ kubectl create namespace <namespace-name>

\$ kubectl create ns my-bank

To switch to a specific namespace: (make this as default type)

\$ kubectl config set-context --current --namespace=<namespace-name>

To list all namespaces:

\$ kubectl get namespaces

To get resources within a specific namespace:

\$ kubectl get <resource-type> -n <namespace-name>

\$ kubectl get deploy -n my-bank

```
$ kubectl get deploy --namespace my-bank
```

```
$ kubectl get all --namespace my-bank
```

```
# To delete a namespace and all associated resources:
```

```
$ kubectl delete namespace <namespace-name>
```

```
$ kubectl delete ns my-bank
```

```
kubectl create ns mydeploy
```

```
kubectl apply -f deploy.yml -n mydeploy
```

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: my-pod
```

```
  namespace: my-demo-ns
```

```
spec:
```

```
  containers:
```

```
    - name: my-container
```

```
      image: nginx:latest
```

```
Normal ScalingReplicaSet 26s deployment-controller Scaled up replica set my-deployment-f47778678 from 0 to 3
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
arvind_2004@ARVINDM:~/my_project$ kubectl create deployment webnginx2 --image=nginx:latest --replicas=1
deployment.apps/webnginx2 created
arvind_2004@ARVINDM:~/my_project$ kubectl replace -f deployment.yml
deployment.apps/my-deployment replaced
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME                                READY   STATUS             RESTARTS   AGE
my-deployment-f47778678-5qspr       0/1     ContainerCreating   0           12s
my-deployment-f47778678-c8d5m       0/1     ErrImagePull        0           12s
my-deployment-f47778678-czpqr       0/1     ImagePullBackOff    0           4m32s
my-deployment-f47778678-d5bxg       0/1     ImagePullBackOff    0           4m32s
my-deployment-f47778678-d8f7j       0/1     ImagePullBackOff    0           4m32s
my-deployment-f47778678-f82gp       0/1     ContainerCreating   0           12s
my-deployment-f47778678-fml6f       0/1     ErrImagePull        0           12s
my-deployment-f47778678-zrh84       0/1     ContainerCreating   0           12s
webnginx2-85b94f8d56-gsz4q         1/1     Running             0           72s
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deploy --replicas=2
error: no objects passed to scale deployments.apps "my-deploy" not found
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy mydeploy --replicas=2
error: no objects passed to scale deployments.apps "mydeploy" not found
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deploy --replicas=2
error: no objects passed to scale deployments.apps "my-deploy" not found
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deployment --replicas=2
deployment.apps/my-deployment scaled
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME                                READY   STATUS             RESTARTS   AGE
my-deployment-f47778678-czpqr       0/1     ImagePullBackOff    0           10m
my-deployment-f47778678-d8f7j       0/1     ImagePullBackOff    0           10m
webnginx2-85b94f8d56-gsz4q         1/1     Running             0           7m13s
arvind_2004@ARVINDM:~/my_project$ kubectl delete webnginx2-85b94f8d56-gsz4q
error: the server doesn't have a resource type "webnginx2-85b94f8d56-gsz4q"
arvind_2004@ARVINDM:~/my_project$ kubectl delete pod webnginx2-85b94f8d56-gsz4q

mypod 1/1 Running 0 82m
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
error: error validating "rs-test.yml": error validating data: apiVersion not set; if you choose to ignore these errors, turn validation off with --validate=false
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
error: error validating "rs-test.yml": error validating data: apiVersion not set; if you choose to ignore these errors, turn validation off with --validate=false
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
replicaset.apps/my-rs created
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
my-app1 0/1 ImagePullBackOff 0 11m
my-pod 1/1 Running 0 90m
my-pod2 1/1 Running 0 89m
my-rs-7xj6 0/1 ContainerCreating 0 8s
my-rs-fd6z6 0/1 ContainerCreating 0 8s
my-rs-h8klx 0/1 ErrImagePull 0 8s
my-rs-wcqzw 0/1 ContainerCreating 0 8s
mypod 1/1 Running 0 93m
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME                                READY   STATUS             RESTARTS   AGE
my-app1 0/1 ImagePullBackOff 0 11m
my-pod 1/1 Running 0 90m
my-pod2 1/1 Running 0 89m
my-rs-7xj6 0/1 ErrImagePull 0 11s
my-rs-fd6z6 0/1 ContainerCreating 0 11s
my-rs-h8klx 0/1 ErrImagePull 0 11s
my-rs-wcqzw 0/1 ContainerCreating 0 11s
mypod 1/1 Running 0 93m
arvind_2004@ARVINDM:~/my_project$ kubectl delete pod my-rs-7xj6
```

```
arvind_2004@ARVINDM: ~/m x + v
4b1990a..a515df1 day4 -> day4
arvind_2004@ARVINDM:~/my_project$ sudo nano my-pod.yml
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME      READY   STATUS    RESTARTS   AGE
my-pod    1/1     Running   0           76m
my-pod2   1/1     Running   0           75m
mypod     1/1     Running   0           79m
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
my-pod    1/1     Running   0           76m
my-pod2   1/1     Running   0           75m
mypod     1/1     Running   0           79m
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
my-pod    1/1     Running   0           77m
my-pod2   1/1     Running   0           76m
mypod     1/1     Running   0           80m
arvind_2004@ARVINDM:~/my_project$ pod run scale
Command 'pod' not found, did you mean:
  command 'od' from deb coreutils (9.4-2ubuntu2)
  command 'pdd' from deb pdd (1.7-1)
  command 'pom' from deb bsdgames (2.17-30)
  command 'pd' from deb puredata-core (0.54.1+ds-3)
  command 'wod' from deb sope-bin (5.9.0-1)
  command 'god' from deb ruby-god (0.13.7-3build5)
  command 'psd' from deb profile-sync-daemon (6.50-1)
  command 'poa' from deb poa (2.0+20060928-8)
  command 'pox' from deb python3-pox (0.3.3-1)
  command 'pwd' from deb coreutils (9.4-2ubuntu2)
  command 'pods' from deb pmtools (2.2.0-3)
  command 'pon' from deb ppp (2.4.9-1+1.1ubuntu1)
  command 'mod' from deb monodoc-base (6.8.0.105+dfsg-3.5ubuntu1)
Try: sudo apt install <deb name>

arvind_2004@ARVINDM: ~/m x + v
/.git: Permission denied
arvind_2004@ARVINDM:/$ kubectl exec -it my-pod2 -- /bin/bash
root@my-pod2:/# ls
bin  dev                docker-entrypoint.sh  home  lib64  mnt  proc  run  srv  tmp  var
boot  docker-entrypoint.d  etc                  lib   media  opt  root  sbin  sys  usr
root@my-pod2:/# exit
exit
arvind_2004@ARVINDM:/$ git init
/.git: Permission denied
arvind_2004@ARVINDM:/$ cd ~/my_project
-bash: cd: /home/arvind_2004/my_project: No such file or directory
arvind_2004@ARVINDM:/$ cd ~
arvind_2004@ARVINDM:~$ mkdir my_project
arvind_2004@ARVINDM:~$ cd my_project
arvind_2004@ARVINDM:~/my_project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/arvind_2004/my_project/.git/
arvind_2004@ARVINDM:~/my_project$ git add pom1.yml
fatal: pathspec 'pom1.yml' did not match any files
arvind_2004@ARVINDM:~/my_project$ ls -l pom1.yml
ls: cannot access 'pom1.yml': No such file or directory
arvind_2004@ARVINDM:~/my_project$ cd /
arvind_2004@ARVINDM:/$ cat pom1.yml
cat: pom1.yml: No such file or directory
```



```
arvind_2004@ARVINDM: ~/m x + v
}
arvind_2004@ARVINDM:/$ sudo nano pod.yml
[sudo] password for arvind_2004:
arvind_2004@ARVINDM:/$ kubectl logs my-pod2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/03/20 04:46:47 [notice] 1#1: using the "epoll" event method
2025/03/20 04:46:47 [notice] 1#1: nginx/1.27.4
2025/03/20 04:46:47 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)
2025/03/20 04:46:47 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2025/03/20 04:46:47 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/03/20 04:46:47 [notice] 1#1: start worker processes
2025/03/20 04:46:47 [notice] 1#1: start worker process 29
2025/03/20 04:46:47 [notice] 1#1: start worker process 30
2025/03/20 04:46:47 [notice] 1#1: start worker process 31
2025/03/20 04:46:47 [notice] 1#1: start worker process 32
2025/03/20 04:46:47 [notice] 1#1: start worker process 33
2025/03/20 04:46:47 [notice] 1#1: start worker process 34
2025/03/20 04:46:47 [notice] 1#1: start worker process 35
2025/03/20 04:46:47 [notice] 1#1: start worker process 36
2025/03/20 04:46:47 [notice] 1#1: start worker process 37
2025/03/20 04:46:47 [notice] 1#1: start worker process 38
2025/03/20 04:46:47 [notice] 1#1: start worker process 39
2025/03/20 04:46:47 [notice] 1#1: start worker process 40
arvind_2004@ARVINDM:/$ kubectl get pod

arvind_2004@ARVINDM: ~/m x + v
arvind_2004@ARVINDM:/$ ls
bin          boot  etc  init  lib.usr-is-merged  lost+found  mnt  proc  run /sbin.usr-is-merged  srv  tmp  var
bin.usr-is-merged  dev  home  lib  lib64  media  opt  root  sbin  snap  sys  usr

arvind_2004@ARVINDM:/$ kubectl get pod -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE       NOMINATED NODE   READINESS GATES
my-pod    1/1     Running   0           3m37s  10.244.0.6    minikube   <none>            <none>
my-pod2   1/1     Running   0           2m38s  10.244.0.7    minikube   <none>            <none>
mypod     1/1     Running   0           6m32s  10.244.0.5    minikube   <none>            <none>

arvind_2004@ARVINDM:/$ kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
coredns-668d6bf9bc-gj2j2            1/1     Running   1 (58m ago)  20h
etcd-minikube                       1/1     Running   1 (58m ago)  20h
kube-apiserver-minikube             1/1     Running   1 (58m ago)  20h
kube-controller-manager-minikube    1/1     Running   1 (58m ago)  20h
kube-proxy-2chv9                    1/1     Running   1 (58m ago)  20h
kube-scheduler-minikube             1/1     Running   1 (58m ago)  20h
storage-provisioner                 1/1     Running   3 (58m ago)  20h

arvind_2004@ARVINDM:/$ kubectl get pods my-pod2 -o yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: "2025-03-20T04:46:43Z"
  labels:
    run: my-pod2
  name: my-pod2
  namespace: default
  resourceVersion: "22354"
  uid: c2ee31e-a019-40aa-b8fb-b5f3b9ddfd3f
spec:
  containers:
  - image: nginx:latest
    imagePullPolicy: Always
    name: my-pod2
```

```
arvind_2004@ARVINDM: ~/m x + v
arvind_2004@ARVINDM:/$ kubectl run my-pod2 --image=nginx:latest --port=8084
pod/my-pod2 created
arvind_2004@ARVINDM:/$ kubectl get pod
NAME      READY   STATUS    RESTARTS   AGE
my-pod    1/1     Running   0           62s
my-pod2   0/1     ContainerCreating   0           3s
mypod     1/1     Running   0           3m57s
arvind_2004@ARVINDM:/$ kubectl logs my-pod2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/03/20 04:46:47 [notice] 1#1: using the "epoll" event method
2025/03/20 04:46:47 [notice] 1#1: nginx/1.27.4
2025/03/20 04:46:47 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)
2025/03/20 04:46:47 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2025/03/20 04:46:47 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/03/20 04:46:47 [notice] 1#1: start worker processes
2025/03/20 04:46:47 [notice] 1#1: start worker process 29
2025/03/20 04:46:47 [notice] 1#1: start worker process 30
2025/03/20 04:46:47 [notice] 1#1: start worker process 31
2025/03/20 04:46:47 [notice] 1#1: start worker process 32
2025/03/20 04:46:47 [notice] 1#1: start worker process 33
2025/03/20 04:46:47 [notice] 1#1: start worker process 34
2025/03/20 04:46:47 [notice] 1#1: start worker process 35
2025/03/20 04:46:47 [notice] 1#1: start worker process 36
2025/03/20 04:46:47 [notice] 1#1: start worker process 37
2025/03/20 04:46:47 [notice] 1#1: start worker process 38

arvind_2004@ARVINDM: ~/m x + v
arvind_2004@ARVINDM:/$ minikube start
🔧 minikube v1.35.0 on Ubuntu 24.04 (amd64)
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.46 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.32.0 on Docker 27.4.1 ...
🔍 Verifying Kubernetes components...
  • Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: default-storageclass, storage-provisioner
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
arvind_2004@ARVINDM:/$ kubectl get pod
No resources found in default namespace.
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx --port=80
pod/mypod created
arvind_2004@ARVINDM:/$ kubectl get pod
NAME      READY   STATUS    RESTARTS   AGE
mypod     1/1     Running   0           73s
arvind_2004@ARVINDM:/$ kubectl get node
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane   19h   v1.32.0
arvind_2004@ARVINDM:/$ kubectl delete pod my-pod
Error from server (NotFound): pods "my-pod" not found
arvind_2004@ARVINDM:/$ kubectl delete pod mypod
pod "mypod" deleted
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx --port=8085
pod/mypod created
arvind_2004@ARVINDM:/$ kubectl get pod
NAME      READY   STATUS    RESTARTS   AGE
mypod     1/1     Running   0           12s
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx:latest --port=8087
Error from server (AlreadyExists): pods "mypod" already exists
arvind_2004@ARVINDM:/$ kubectl run my-pod --image=nginx:latest --port=8087
```

```
arvind_2004@ARVINDM: ~/m x + v
arvind_2004@ARVINDM:~/my_project$ kubectl delete all --all
pod "my-deploy-6b478cb7cb-4fhwv" deleted
service "kubernetes" deleted
service "my-service" deleted
deployment.apps "my-deploy" deleted
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
deployment.apps/my-deploy created
service/my-service created
arvind_2004@ARVINDM:~/my_project$ minikube service my-service

```

NAMESPACE	NAME	TARGET PORT	URL
default	my-service	9001	http://192.168.49.2:30005

```

★ Starting tunnel for service my-service.

```

NAMESPACE	NAME	TARGET PORT	URL
default	my-service		http://127.0.0.1:36227

```

🔗 Opening service default/my-service in default browser...
🔗 http://127.0.0.1:36227
! Because you are using a Docker driver on linux, the terminal needs to be open to run it.
^C 🛑 Stopping tunnel for service my-service.
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005
<!doctype html><html lang="en"><head><title>HTTP Status 404 - Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,
sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {
font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 -
Not Found</h1><hr class="line" /><p><b>Type</b></p><p><b>Description</b></p><b>The origin server did not find a current repre
sentation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3>
</body></html>arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-app/
<!doctype html><html lang="en"><head><title>HTTP Status 404 - Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,
sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {
font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 -
Not Found</h1><hr class="line" /><p><b>Type</b></p><p><b>Description</b></p><b>The origin server did not find a current repre
sentation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3>
</body></html>arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/maven-web-app/
<!doctype html><html lang="en"><head><title>HTTP Status 404 - Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,
sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {
font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 -
Not Found</h1><hr class="line" /><p><b>Type</b></p><p><b>Description</b></p><b>The origin server did not find a current repre
sentation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3>
</body></html>arvind_2004@ARVINDM:~/my_project$ kubectl exec -it my-deploy-6b478cb7cb-t5lkg -- /bin/bash
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat# ls
bin          conf          filtered-KEYS  LICENSE  native-jni-lib  README.md  RUNNING.txt  upstream-KEYS  webapps.dist
BUILDING.txt  CONTRIBUTING.md  lib           Logs     NOTICE         RELEASE-NOTES  temp         webapps       work
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat# cd webapps
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat/webapps# ls
my-web  my-web.war
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat/webapps# cd
root@my-deploy-6b478cb7cb-t5lkg:~# #
exit
arvind_2004@ARVINDM:~/my_project$ ls
deployment.yml  kubctl.txt  my-pod.yml  rs-test.yml
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-web/
<html>
<body>
<h2>Hello World!</h2>
</body>
</html>
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-web/
<html>
<body>
<h2>Hello World!</h2>
</body>
</html>
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-web/
```

```
arvind_2004@ARVINDM: ~/m x + v
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
service/my-service created
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
my-deploy-5df65497db-xfhfq          0/1     ErrImagePull    0           3m40s
my-deployment-f47778678-czpqz       0/1     ImagePullBackOff  0           41m
my-deployment-f47778678-d8f7j       0/1     ImagePullBackOff  0           41m
webnginx2-85b94f8d56-cvxv2         1/1     Running         0           29m
arvind_2004@ARVINDM:~/my_project$ kubectl get services
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes  ClusterIP   10.96.0.1    <none>        443/TCP          44m
my-service  NodePort    10.106.84.41 <none>        7070:30002/TCP   31s
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
[sudo] password for arvind_2004:
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
deployment.apps/my-deploy configured
service/my-service unchanged
arvind_2004@ARVINDM:~/my_project$ kubectl replace deployment.yml
error: must specify one of -f and -k
arvind_2004@ARVINDM:~/my_project$ kubectl replace -f deployment.yml
deployment.apps/my-deploy replaced
service/my-service replaced
arvind_2004@ARVINDM:~/my_project$ minikube service my-service
-----
| NAMESPACE | NAME       | TARGET PORT | URL                               |
|-----|-----|-----|-----|
| default   | my-service | 7070        | http://192.168.49.2:30002       |
|-----|-----|-----|-----|
✖ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found
```

apiVersion: v1

kind: Pod

metadata:

name: my-pod2

spec:

containers:

- name: arvindm2004/webapp1

image: arvindm2004/webapp1:tagname

ports:

- containerPort: 8084

1. Create a pod using run command

\$ kubectl run <pod-name> --image=<image-name> --port=<container-port>

\$ kubectl run my-pod --image=nginx --port=80

2. View all the pods

(In default namespace)

```
$ kubectl get pods
```

(In All namespace)

```
$ kubectl get pods -A
```

For a specific namespace

```
$ kubectl get pods -n kube-system
```

For a specific type

```
$ kubectl get pods <pod-name>
```

```
$ kubectl get pods <pod-name> -o wide
```

```
$ kubectl get pods <pod-name> -o yaml
```

```
$ kubectl get pods <pod-name> -o json
```

3. Describe a pod (View Pod details)

```
$ kubectl describe pod <pod-name>
```

```
$ kubectl describe pod my-pod
```

4. View Logs of a pod

```
$ kubectl logs <pod-name>
```

```
$ kubectl logs my-pod
```

5. Execute any command inside Pod (Inside Pod OS)

```
$ kubectl exec <pod-name> -- <command>
```

CMD:

Microsoft Windows [Version 10.0.22631.4751]

(c) Microsoft Corporation. All rights reserved.

C:\Users\Arvind M>wsl.exe -d Ubuntu

Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/pro>

System information as of Thu Mar 20 04:31:21 UTC 2025

System load: 0.83 Processes: 36

Usage of /: 0.8% of 1006.85GB Users logged in: 0

Memory usage: 12% IPv4 address for eth0: 172.27.90.197

Swap usage: 0%

This message is shown once a day. To disable it please create the
/home/arvind_2004/.hushlogin file.

arvind_2004@ARVINDM:/mnt/c/Users/Arvind M\$ cd ..

arvind_2004@ARVINDM:/mnt/c/Users\$ cd /

arvind_2004@ARVINDM:/\$ minikube start

😊 minikube v1.35.0 on Ubuntu 24.04 (amd64)

🔧 Using the docker driver based on existing profile

👍 Starting "minikube" primary control-plane node in "minikube" cluster

 Pulling base image v0.0.46 ...

 Restarting existing docker container for "minikube" ...

 Preparing Kubernetes v1.32.0 on Docker 27.4.1 ...

 Verifying Kubernetes components...

- Using image gcr.io/k8s-minikube/storage-provisioner:v5

 Enabled addons: default-storageclass, storage-provisioner

 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

```
arvind_2004@ARVINDM:/$ kubectl get pod
```

No resources found in default namespace.

```
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx --port=80
```

pod/mypod created

```
arvind_2004@ARVINDM:/$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

mypod	1/1	Running	0	73s
-------	-----	---------	---	-----

```
arvind_2004@ARVINDM:/$ kubectl get node
```

NAME	STATUS	ROLES	AGE	VERSION
------	--------	-------	-----	---------

minikube	Ready	control-plane	19h	v1.32.0
----------	-------	---------------	-----	---------

```
arvind_2004@ARVINDM:/$ kubectl delete pod my-pod
```

Error from server (NotFound): pods "my-pod" not found

```
arvind_2004@ARVINDM:/$ kubectl delete pod mypod
```

pod "mypod" deleted

```
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx --port=8085
```

pod/mypod created

```
arvind_2004@ARVINDM:/$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

mypod	1/1	Running	0	12s
-------	-----	---------	---	-----

```
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx:latest --port=8087
```

Error from server (AlreadyExists): pods "mypod" already exists

```
arvind_2004@ARVINDM:/$ kubectl run my-pod --image=nginx:latest --port=8087
```

pod/my-pod created

```
arvind_2004@ARVINDM:/$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-pod	1/1	Running	0	8s
--------	-----	---------	---	----

my-pod	1/1	Running	0	3m3s
--------	-----	---------	---	------

```
arvind_2004@ARVINDM:/$ kubectl run my-pod2 --image=nginx:latest --port=8084
```

pod/my-pod2 created

```
arvind_2004@ARVINDM:/$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-pod	1/1	Running	0	62s
--------	-----	---------	---	-----

my-pod2	0/1	ContainerCreating	0	3s
---------	-----	-------------------	---	----

my-pod	1/1	Running	0	3m57s
--------	-----	---------	---	-------

```
arvind_2004@ARVINDM:/$ kubectl logs my-pod2
```

/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration

/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/

/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh

10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf

10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf

/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh

/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh

/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh

/docker-entrypoint.sh: Configuration complete; ready for start up

2025/03/20 04:46:47 [notice] 1#1: using the "epoll" event method

2025/03/20 04:46:47 [notice] 1#1: nginx/1.27.4

2025/03/20 04:46:47 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)

2025/03/20 04:46:47 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2

2025/03/20 04:46:47 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576

2025/03/20 04:46:47 [notice] 1#1: start worker processes

2025/03/20 04:46:47 [notice] 1#1: start worker process 29

2025/03/20 04:46:47 [notice] 1#1: start worker process 30

2025/03/20 04:46:47 [notice] 1#1: start worker process 31

2025/03/20 04:46:47 [notice] 1#1: start worker process 32

2025/03/20 04:46:47 [notice] 1#1: start worker process 33

2025/03/20 04:46:47 [notice] 1#1: start worker process 34

2025/03/20 04:46:47 [notice] 1#1: start worker process 35

2025/03/20 04:46:47 [notice] 1#1: start worker process 36

2025/03/20 04:46:47 [notice] 1#1: start worker process 37

2025/03/20 04:46:47 [notice] 1#1: start worker process 38

2025/03/20 04:46:47 [notice] 1#1: start worker process 39

2025/03/20 04:46:47 [notice] 1#1: start worker process 40

arvind_2004@ARVINDM:/ \$ minikube ssh

docker@minikube:~\$ docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
2c3ef2e4ea91	nginx	"/docker-entrypoint...."	About a minute ago	Up About a minute
k8s_my-pod2_my-pod2_default_c2eee31e-a019-40aa-b8fb-b5f3b9ddfd3f_0				
3c85de85a4ac	registry.k8s.io/pause:3.10	"/pause"	About a minute ago	Up About a minute
k8s_POD_my-pod2_default_c2eee31e-a019-40aa-b8fb-b5f3b9ddfd3f_0				
18fefc6bc8ad	nginx	"/docker-entrypoint...."	2 minutes ago	Up 2 minutes
k8s_my-pod_my-pod_default_c76700d2-2a03-4cd2-8887-9adc28ee0e72_0				
1961b345922d	registry.k8s.io/pause:3.10	"/pause"	2 minutes ago	Up 2 minutes
k8s_POD_my-pod_default_c76700d2-2a03-4cd2-8887-9adc28ee0e72_0				
43ca4ffd4f14	nginx	"/docker-entrypoint...."	5 minutes ago	Up 5 minutes
k8s_mypod_mypod_default_8671366b-5ac7-4c55-9a4a-40a746680ba7_0				

a81a4eab7452	registry.k8s.io/pause:3.10	"/pause"	5 minutes ago	Up 5 minutes
k8s_POD_mypod_default_8671366b-5ac7-4c55-9a4a-40a746680ba7_0				
68039fa229f4	6e38f40d628d	"/storage-provisioner"	16 minutes ago	Up 16 minutes
k8s_storage-provisioner_storage-provisioner_kube-system_662c7ba5-f633-49b7-8f08-8fe775a520ee_3				
c2f36b460128	c69fa2e9cbf5	"/coredns -conf /etc..."	16 minutes ago	Up 16 minutes
k8s_coredns_coredns-668d6bf9bc-gj2j2_kube-system_7bfcff89-694f-406a-808e-466d0d549ea8_1				
20035d5be14e	040f9f8aac8c	"/usr/local/bin/kube..."	16 minutes ago	Up 16 minutes
k8s_kube-proxy_kube-proxy-2chv9_kube-system_b55c713f-8f1e-4218-b286-07528e7c34ca_1				
45f42d177058	registry.k8s.io/pause:3.10	"/pause"	16 minutes ago	Up 16 minutes
k8s_POD_coredns-668d6bf9bc-gj2j2_kube-system_7bfcff89-694f-406a-808e-466d0d549ea8_1				
1124f740e47b	registry.k8s.io/pause:3.10	"/pause"	16 minutes ago	Up 16 minutes
k8s_POD_storage-provisioner_kube-system_662c7ba5-f633-49b7-8f08-8fe775a520ee_1				
3bb28a2ca964	registry.k8s.io/pause:3.10	"/pause"	16 minutes ago	Up 16 minutes
k8s_POD_kube-proxy-2chv9_kube-system_b55c713f-8f1e-4218-b286-07528e7c34ca_1				
13ad1eb82586	a9e7e6b294ba	"etcd --advertise-cl..."	16 minutes ago	Up 16 minutes
k8s_etcd_etcd-minikube_kube-system_2b4b75c2a289008e0b381891e9683040_1				
9c99aea7da36	c2e17b8d0f4a	"kube-apiserver --ad..."	16 minutes ago	Up 16 minutes
k8s_kube-apiserver_kube-apiserver-minikube_kube-system_d72d0a4cf4be077c9919d46b7358a5e8_1				
09bb410b3d82	8cab3d2a8bd0	"kube-controller-man..."	16 minutes ago	Up 16 minutes
k8s_kube-controller-manager_kube-controller-manager-minikube_kube-system_843c74f7b3bc7d7040a05c31708a6a30_1				
cae7502e8a41	a389e107f4ff	"kube-scheduler --au..."	16 minutes ago	Up 16 minutes
k8s_kube-scheduler_kube-scheduler-minikube_kube-system_d14ce008bee3a1f3bd7cf547688f9dfe_1				
80138199261a	registry.k8s.io/pause:3.10	"/pause"	16 minutes ago	Up 16 minutes
k8s_POD_kube-apiserver-minikube_kube-system_d72d0a4cf4be077c9919d46b7358a5e8_1				

a0e780589b00 registry.k8s.io/pause:3.10 "/pause" 16 minutes ago Up 16 minutes
k8s_POD_etcd-minikube_kube-system_2b4b75c2a289008e0b381891e9683040_1

4acf7db67313 registry.k8s.io/pause:3.10 "/pause" 16 minutes ago Up 16 minutes
k8s_POD_kube-controller-manager-minikube_kube-system_843c74f7b3bc7d7040a05c31708a6a30_1

a00a5c570e77 registry.k8s.io/pause:3.10 "/pause" 16 minutes ago Up 16 minutes
k8s_POD_kube-scheduler-minikube_kube-system_d14ce008bee3a1f3bd7cf547688f9dfe_1

docker@minikube:~\$ exit

logout

arvind_2004@ARVINDM:/\$ ls

bin boot etc init lib.usr-is-merged lost+found mnt proc run sbin.usr-is-merged
srv tmp var

bin.usr-is-merged dev home lib lib64 media opt root sbin snap sys
usr

arvind_2004@ARVINDM:/\$ kubectl get pod -o wide

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED	NODE
my-pod	1/1	Running	0	3m37s	10.244.0.6	minikube	<none>	<none>
my-pod2	1/1	Running	0	2m38s	10.244.0.7	minikube	<none>	<none>
mypod	1/1	Running	0	6m32s	10.244.0.5	minikube	<none>	<none>

arvind_2004@ARVINDM:/\$ kubectl get pods -n kube-system

NAME	READY	STATUS	RESTARTS	AGE
coredns-668d6bf9bc-gj2j2	1/1	Running	1 (58m ago)	20h
etcd-minikube	1/1	Running	1 (58m ago)	20h
kube-apiserver-minikube	1/1	Running	1 (58m ago)	20h
kube-controller-manager-minikube	1/1	Running	1 (58m ago)	20h
kube-proxy-2chv9	1/1	Running	1 (58m ago)	20h
kube-scheduler-minikube	1/1	Running	1 (58m ago)	20h
storage-provisioner	1/1	Running	3 (58m ago)	20h

```
arvind_2004@ARVINDM:/ $ kubectl get pods my-pod2 -o yaml
```

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  creationTimestamp: "2025-03-20T04:46:43Z"
```

```
  labels:
```

```
    run: my-pod2
```

```
  name: my-pod2
```

```
  namespace: default
```

```
  resourceVersion: "22354"
```

```
  uid: c2eee31e-a019-40aa-b8fb-b5f3b9ddfd3f
```

```
spec:
```

```
  containers:
```

```
    - image: nginx:latest
```

```
      imagePullPolicy: Always
```

```
      name: my-pod2
```

```
      ports:
```

```
        - containerPort: 8084
```

```
          protocol: TCP
```

```
      resources: {}
```

```
      terminationMessagePath: /dev/termination-log
```

```
      terminationMessagePolicy: File
```

```
      volumeMounts:
```

```
        - mountPath: /var/run/secrets/kubernetes.io/serviceaccount
```

```
          name: kube-api-access-ff8h2
```

```
          readOnly: true
```

```
  dnsPolicy: ClusterFirst
```

```
  enableServiceLinks: true
```

nodeName: minikube

preemptionPolicy: PreemptLowerPriority

priority: 0

restartPolicy: Always

schedulerName: default-scheduler

securityContext: {}

serviceAccount: default

serviceAccountName: default

terminationGracePeriodSeconds: 30

tolerations:

- effect: NoExecute
- key: node.kubernetes.io/not-ready
- operator: Exists
- tolerationSeconds: 300
- effect: NoExecute
- key: node.kubernetes.io/unreachable
- operator: Exists
- tolerationSeconds: 300

volumes:

- name: kube-api-access-ff8h2
- projected:
 - defaultMode: 420
 - sources:
 - serviceAccountToken:
 - expirationSeconds: 3607
 - path: token
- configMap:
 - items:

- key: ca.crt
 - path: ca.crt
 - name: kube-root-ca.crt
- downwardAPI:
 - items:
 - fieldRef:
 - apiVersion: v1
 - fieldPath: metadata.namespace
 - path: namespace

status:

conditions:

- lastProbeTime: null
 - lastTransitionTime: "2025-03-20T04:46:47Z"
 - status: "True"
 - type: PodReadyToStartContainers
- lastProbeTime: null
 - lastTransitionTime: "2025-03-20T04:46:43Z"
 - status: "True"
 - type: Initialized
- lastProbeTime: null
 - lastTransitionTime: "2025-03-20T04:46:47Z"
 - status: "True"
 - type: Ready
- lastProbeTime: null
 - lastTransitionTime: "2025-03-20T04:46:47Z"
 - status: "True"
 - type: ContainersReady
- lastProbeTime: null

lastTransitionTime: "2025-03-20T04:46:43Z"

status: "True"

type: PodScheduled

containerStatuses:

- containerID:

docker://2c3ef2e4ea91ddddd6ada681eab9430a4888c1c17351df593c32dd7ada2f8c06b

image: nginx:latest

imageID: docker-

pullable://nginx@sha256:124b44bfc9ccd1f3cedf4b592d4d1e8bddb78b51ec2ed5056c52d3692baebc19

lastState: {}

name: my-pod2

ready: true

restartCount: 0

started: true

state:

running:

startedAt: "2025-03-20T04:46:47Z"

volumeMounts:

- mountPath: /var/run/secrets/kubernetes.io/serviceaccount

name: kube-api-access-ff8h2

readOnly: true

recursiveReadOnly: Disabled

hostIP: 192.168.49.2

hostIPs:

- ip: 192.168.49.2

phase: Running

podIP: 10.244.0.7

podIPs:

- ip: 10.244.0.7

qosClass: BestEffort

startTime: "2025-03-20T04:46:43Z"

arvind_2004@ARVINDM:/\$ kubectl get pods my-pod2 -o json

```
{
  "apiVersion": "v1",
  "kind": "Pod",
  "metadata": {
    "creationTimestamp": "2025-03-20T04:46:43Z",
    "labels": {
      "run": "my-pod2"
    },
    "name": "my-pod2",
    "namespace": "default",
    "resourceVersion": "22354",
    "uid": "c2eee31e-a019-40aa-b8fb-b5f3b9ddfd3f"
  },
  "spec": {
    "containers": [
      {
        "image": "nginx:latest",
        "imagePullPolicy": "Always",
        "name": "my-pod2",
        "ports": [
          {
            "containerPort": 8084,
            "protocol": "TCP"
          }
        ]
      }
    ]
  }
}
```



```
    ],  
    "resources": {},  
    "terminationMessagePath": "/dev/termination-log",  
    "terminationMessagePolicy": "File",  
    "volumeMounts": [  
      {  
        "mountPath": "/var/run/secrets/kubernetes.io/serviceaccount",  
        "name": "kube-api-access-ff8h2",  
        "readOnly": true  
      }  
    ]  
  }  
],  
  "dnsPolicy": "ClusterFirst",  
  "enableServiceLinks": true,  
  "nodeName": "minikube",  
  "preemptionPolicy": "PreemptLowerPriority",  
  "priority": 0,  
  "restartPolicy": "Always",  
  "schedulerName": "default-scheduler",  
  "securityContext": {},  
  "serviceAccount": "default",  
  "serviceAccountName": "default",  
  "terminationGracePeriodSeconds": 30,  
  "tolerations": [  
    {  
      "effect": "NoExecute",  
      "key": "node.kubernetes.io/not-ready",
```

```
    "operator": "Exists",
    "tolerationSeconds": 300
  },
  {
    "effect": "NoExecute",
    "key": "node.kubernetes.io/unreachable",
    "operator": "Exists",
    "tolerationSeconds": 300
  }
],
"volumes": [
  {
    "name": "kube-api-access-ff8h2",
    "projected": {
      "defaultMode": 420,
      "sources": [
        {
          "serviceAccountToken": {
            "expirationSeconds": 3607,
            "path": "token"
          }
        },
        {
          "configMap": {
            "items": [
              {
                "key": "ca.crt",
                "path": "ca.crt"
              }
            ]
          }
        }
      ]
    }
  }
]
```

```
        }
      ],
      "name": "kube-root-ca.crt"
    }
  },
  {
    "downwardAPI": {
      "items": [
        {
          "fieldRef": {
            "apiVersion": "v1",
            "fieldPath": "metadata.namespace"
          },
          "path": "namespace"
        }
      ]
    }
  }
]
},
"status": {
  "conditions": [
    {
      "lastProbeTime": null,
      "lastTransitionTime": "2025-03-20T04:46:47Z",
```

```
    "status": "True",
    "type": "PodReadyToStartContainers"
  },
  {
    "lastProbeTime": null,
    "lastTransitionTime": "2025-03-20T04:46:43Z",
    "status": "True",
    "type": "Initialized"
  },
  {
    "lastProbeTime": null,
    "lastTransitionTime": "2025-03-20T04:46:47Z",
    "status": "True",
    "type": "Ready"
  },
  {
    "lastProbeTime": null,
    "lastTransitionTime": "2025-03-20T04:46:47Z",
    "status": "True",
    "type": "ContainersReady"
  },
  {
    "lastProbeTime": null,
    "lastTransitionTime": "2025-03-20T04:46:43Z",
    "status": "True",
    "type": "PodScheduled"
  }
],
```

```
"containerStatuses": [  
  {  
    "containerID":  
"docker://2c3ef2e4ea91dddd6ada681eab9430a4888c1c17351df593c32dd7ada2f8c06b",  
    "image": "nginx:latest",  
    "imageID": "docker-  
pullable://nginx@sha256:124b44bfc9ccd1f3cedf4b592d4d1e8bddb78b51ec2ed5056c52d36  
92baebc19",  
    "lastState": {},  
    "name": "my-pod2",  
    "ready": true,  
    "restartCount": 0,  
    "started": true,  
    "state": {  
      "running": {  
        "startedAt": "2025-03-20T04:46:47Z"  
      }  
    },  
    "volumeMounts": [  
      {  
        "mountPath": "/var/run/secrets/kubernetes.io/serviceaccount",  
        "name": "kube-api-access-ff8h2",  
        "readOnly": true,  
        "recursiveReadOnly": "Disabled"  
      }  
    ]  
  }  
],  
  "hostIP": "192.168.49.2",
```

```
"hostIPs": [  
  {  
    "ip": "192.168.49.2"  
  }  
,  
  "phase": "Running",  
  "podIP": "10.244.0.7",  
  "podIPs": [  
    {  
      "ip": "10.244.0.7"  
    }  
,  
    "qosClass": "BestEffort",  
    "startTime": "2025-03-20T04:46:43Z"  
  ]  
}
```

```
arvind_2004@ARVINDM:/$ sudo nano pod.yml
```

```
[sudo] password for arvind_2004:
```

```
arvind_2004@ARVINDM:/$ kubectl logs my-pod2
```

```
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform  
configuration
```

```
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
```

```
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
```

```
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
```

```
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
```

```
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
```

```
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
```

```
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
```

```
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/03/20 04:46:47 [notice] 1#1: using the "epoll" event method
2025/03/20 04:46:47 [notice] 1#1: nginx/1.27.4
2025/03/20 04:46:47 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)
2025/03/20 04:46:47 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2025/03/20 04:46:47 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/03/20 04:46:47 [notice] 1#1: start worker processes
2025/03/20 04:46:47 [notice] 1#1: start worker process 29
2025/03/20 04:46:47 [notice] 1#1: start worker process 30
2025/03/20 04:46:47 [notice] 1#1: start worker process 31
2025/03/20 04:46:47 [notice] 1#1: start worker process 32
2025/03/20 04:46:47 [notice] 1#1: start worker process 33
2025/03/20 04:46:47 [notice] 1#1: start worker process 34
2025/03/20 04:46:47 [notice] 1#1: start worker process 35
2025/03/20 04:46:47 [notice] 1#1: start worker process 36
2025/03/20 04:46:47 [notice] 1#1: start worker process 37
2025/03/20 04:46:47 [notice] 1#1: start worker process 38
2025/03/20 04:46:47 [notice] 1#1: start worker process 39
2025/03/20 04:46:47 [notice] 1#1: start worker process 40
```

```
arvind_2004@ARVINDM:/$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
my-pod	1/1	Running	0	54m
my-pod2	1/1	Running	0	53m
mypod	1/1	Running	0	57m

```
arvind_2004@ARVINDM:/$ sudo nano pod1.yml
```

```
arvind_2004@ARVINDM:/$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
my-pod	1/1	Running	0	57m

```
my-pod2 1/1 Running 0 56m
mypod 1/1 Running 0 60m
arvind_2004@ARVINDM:/$ kubectl exec -it my-pod2 -- /bin/bash
root@my-pod2:/# exit
exit
arvind_2004@ARVINDM:/$ git init
/.git: Permission denied
arvind_2004@ARVINDM:/$ kubectl exec -it my-pod2 -- /bin/bash
root@my-pod2:/# ls
bin dev docker-entrypoint.sh home lib64 mnt proc run srv tmp var
boot docker-entrypoint.d etc lib media opt root sbin sys usr
root@my-pod2:/# exit
exit
arvind_2004@ARVINDM:/$ git init
/.git: Permission denied
arvind_2004@ARVINDM:/$ cd ~/my_project
-bash: cd: /home/arvind_2004/my_project: No such file or directory
arvind_2004@ARVINDM:/$ cd ~
arvind_2004@ARVINDM:~$ mkdir my_project
arvind_2004@ARVINDM:~$ cd my_project
arvind_2004@ARVINDM:~/my_project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint: git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
```


hint: 'development'. The just-created branch can be renamed via this command:

hint:

hint: git branch -m <name>

Initialized empty Git repository in /home/arvind_2004/my_project/.git/

arvind_2004@ARVINDM:~/my_project\$ git add pom1.yml

fatal: pathspec 'pom1.yml' did not match any files

arvind_2004@ARVINDM:~/my_project\$ ls -l pom1.yml

ls: cannot access 'pom1.yml': No such file or directory

arvind_2004@ARVINDM:~/my_project\$ cd /

arvind_2004@ARVINDM:/\$ cat pom1.yml

cat: pom1.yml: No such file or directory

arvind_2004@ARVINDM:/\$ cat pod1.yml

apiVersion: v1

kind: Pod

metadata:

name: my-pod2

spec:

containers:

- name: arvindm2004/webapp1

image: arvindm2004/webapp1:tagname

ports:

- containerPort: 8084

arvind_2004@ARVINDM:/\$ cd ~/my_project

arvind_2004@ARVINDM:~/my_project\$ sudo nano pod2.yml

arvind_2004@ARVINDM:~/my_project\$ git init

Reinitialized existing Git repository in /home/arvind_2004/my_project/.git/

arvind_2004@ARVINDM:~/my_project\$ git add .

arvind_2004@ARVINDM:~/my_project\$ git commit -m "pod"

```
[master (root-commit) 9e611cd] pod
1 file changed, 10 insertions(+)
create mode 100644 pod2.yml

arvind_2004@ARVINDM:~/my_project$ git remote add origin
https://github.com/Arvind22cse/devops_tasks.git

arvind_2004@ARVINDM:~/my_project$ git checkout day4
error: pathspec 'day4' did not match any file(s) known to git

arvind_2004@ARVINDM:~/my_project$ git switch day4
fatal: invalid reference: day4

arvind_2004@ARVINDM:~/my_project$ git branch day4

arvind_2004@ARVINDM:~/my_project$ git checkout day4
Switched to branch 'day4'

arvind_2004@ARVINDM:~/my_project$ git push origin day4
Username for 'https://github.com': Arvind22cse
Password for 'https://Arvind22cse@github.com':
To https://github.com/Arvind22cse/devops_tasks.git
! [rejected]    day4 -> day4 (fetch first)
error: failed to push some refs to 'https://github.com/Arvind22cse/devops_tasks.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

arvind_2004@ARVINDM:~/my_project$ git pull
remote: Enumerating objects: 29, done.
remote: Counting objects: 100% (29/29), done.
remote: Compressing objects: 100% (27/27), done.
remote: Total 29 (delta 0), reused 29 (delta 0), pack-reused 0 (from 0)
```

Unpacking objects: 100% (29/29), 3.83 MiB | 2.70 MiB/s, done.

From https://github.com/Arvind22cse/devops_tasks

* [new branch] day1 -> origin/day1

* [new branch] day2 -> origin/day2

* [new branch] day3 -> origin/day3

* [new branch] day4 -> origin/day4

There is no tracking information for the current branch.

Please specify which branch you want to merge with.

See `git-pull(1)` for details.

```
git pull <remote> <branch>
```

If you wish to set tracking information for this branch you can do so with:

```
git branch --set-upstream-to=origin/<branch> day4
```

```
arvind_2004@ARVINDM:~/my_project$ git add .
```

```
arvind_2004@ARVINDM:~/my_project$ git commit -m "pod"
```

On branch day4

nothing to commit, working tree clean

```
arvind_2004@ARVINDM:~/my_project$ git push origin day4
```

Username for 'https://github.com': Arvind22cse

Password for 'https://Arvind22cse@github.com':

To https://github.com/Arvind22cse/devops_tasks.git

! [rejected] day4 -> day4 (non-fast-forward)

error: failed to push some refs to 'https://github.com/Arvind22cse/devops_tasks.git'

hint: Updates were rejected because the tip of your current branch is behind

hint: its remote counterpart. If you want to integrate the remote changes,

hint: use 'git pull' before pushing again.

hint: See the 'Note about fast-forwards' in 'git push --help' for details.

```
arvind_2004@ARVINDM:~/my_project$ git commit -m "pod"
```

On branch day4

nothing to commit, working tree clean

```
arvind_2004@ARVINDM:~/my_project$ git push origin day4
```

Username for 'https://github.com': ^C

```
arvind_2004@ARVINDM:~/my_project$ git pull --rebase origin day4
```

From https://github.com/Arvind22cse/devops_tasks

```
* branch      day4    -> FETCH_HEAD
```

Successfully rebased and updated refs/heads/day4.

```
arvind_2004@ARVINDM:~/my_project$ git add .
```

```
arvind_2004@ARVINDM:~/my_project$ git rebase --continue
```

fatal: No rebase in progress?

```
arvind_2004@ARVINDM:~/my_project$ git push origin day4 --force-with-lease
```

Username for 'https://github.com': Arvind22cse

Password for 'https://Arvind22cse@github.com':

Enumerating objects: 4, done.

Counting objects: 100% (4/4), done.

Delta compression using up to 12 threads

Compressing objects: 100% (3/3), done.

Writing objects: 100% (3/3), 397 bytes | 397.00 KiB/s, done.

Total 3 (delta 0), reused 0 (delta 0), pack-reused 0

To https://github.com/Arvind22cse/devops_tasks.git

```
4b1990a..a515df1 day4 -> day4
```

```
arvind_2004@ARVINDM:~/my_project$ sudo nano my-pod.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-pod 1/1 Running 0 76m

my-pod2 1/1 Running 0 75m

mypod 1/1 Running 0 79m

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-pod	1/1	Running	0	76m
--------	-----	---------	---	-----

my-pod2	1/1	Running	0	75m
---------	-----	---------	---	-----

mypod	1/1	Running	0	79m
-------	-----	---------	---	-----

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-pod	1/1	Running	0	77m
--------	-----	---------	---	-----

my-pod2	1/1	Running	0	76m
---------	-----	---------	---	-----

mypod	1/1	Running	0	80m
-------	-----	---------	---	-----

arvind_2004@ARVINDM:~/my_project\$ pod run scale

Command 'pod' not found, did you mean:

command 'od' from deb coreutils (9.4-2ubuntu2)

command 'pdd' from deb pdd (1.7-1)

command 'pom' from deb bsdgames (2.17-30)

command 'pd' from deb puredata-core (0.54.1+ds-3)

command 'wod' from deb sope-bin (5.9.0-1)

command 'god' from deb ruby-god (0.13.7-3build5)

command 'psd' from deb profile-sync-daemon (6.50-1)

command 'poa' from deb poa (2.0+20060928-8)

command 'pox' from deb python3-pox (0.3.3-1)

command 'pwd' from deb coreutils (9.4-2ubuntu2)

command 'pods' from deb pmttools (2.2.0-3)

command 'pon' from deb ppp (2.4.9-1+1.1ubuntu1)

command 'mod' from deb monodoc-base (6.8.0.105+dfsg-3.5ubuntu1)

Try: `sudo apt install <deb name>`

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f my-pod.yml
```

pod/my-app1 created

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-app1	0/1	ErrImagePull	0	13s
---------	-----	--------------	---	-----

my-pod	1/1	Running	0	79m
--------	-----	---------	---	-----

my-pod2	1/1	Running	0	78m
---------	-----	---------	---	-----

mypod	1/1	Running	0	82m
-------	-----	---------	---	-----

```
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
```

error: error validating "rs-test.yml": error validating data: apiVersion not set; if you choose to ignore these errors, turn validation off with --validate=false

```
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
```

error: error validating "rs-test.yml": error validating data: apiVersion not set; if you choose to ignore these errors, turn validation off with --validate=false

```
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
```

replicaset.apps/my-rs created

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-app1	0/1	ImagePullBackOff	0	11m
---------	-----	------------------	---	-----

my-pod	1/1	Running	0	90m
--------	-----	---------	---	-----

my-pod2	1/1	Running	0	89m
---------	-----	---------	---	-----

my-rs-7xjf6	0/1	ContainerCreating	0	8s
-------------	-----	-------------------	---	----

my-rs-fd6z6	0/1	ContainerCreating	0	8s
-------------	-----	-------------------	---	----

my-rs-h8klx	0/1	ErrImagePull	0	8s
-------------	-----	--------------	---	----

my-rs-wcqzw	0/1	ContainerCreating	0	8s
-------------	-----	-------------------	---	----

```
mypod    1/1    Running      0    93m
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-app1	0/1	ImagePullBackOff	0	11m
---------	-----	------------------	---	-----

my-pod	1/1	Running	0	90m
--------	-----	---------	---	-----

my-pod2	1/1	Running	0	89m
---------	-----	---------	---	-----

my-rs-7xjf6	0/1	ErrImagePull	0	11s
-------------	-----	--------------	---	-----

my-rs-fd6z6	0/1	ContainerCreating	0	11s
-------------	-----	-------------------	---	-----

my-rs-h8klx	0/1	ErrImagePull	0	11s
-------------	-----	--------------	---	-----

my-rs-wcqzw	0/1	ContainerCreating	0	11s
-------------	-----	-------------------	---	-----

```
mypod    1/1    Running      0    93m
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl delete pod my-rs-7xjf6
```

```
pod "my-rs-7xjf6" deleted
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-app1	0/1	ImagePullBackOff	0	12m
---------	-----	------------------	---	-----

my-pod	1/1	Running	0	91m
--------	-----	---------	---	-----

my-pod2	1/1	Running	0	90m
---------	-----	---------	---	-----

my-rs-ctmbx	0/1	ContainerCreating	0	3s
-------------	-----	-------------------	---	----

my-rs-fd6z6	0/1	ImagePullBackOff	0	66s
-------------	-----	------------------	---	-----

my-rs-h8klx	0/1	ImagePullBackOff	0	66s
-------------	-----	------------------	---	-----

my-rs-wcqzw	0/1	ErrImagePull	0	66s
-------------	-----	--------------	---	-----

```
mypod    1/1    Running      0    94m
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl exec -it my-rs-ctmbx -- /bin/bash
```

```
error: Internal error occurred: unable to upgrade connection: container not found ("my-app")
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl describe my-rs-ctmbx
```

```
error: the server doesn't have a resource type "my-rs-ctmbx"
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl describe my-rs-fd6z6
```

```
error: the server doesn't have a resource type "my-rs-fd6z6"
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
my-app1	0/1	ImagePullBackOff	0	138m
my-pod	1/1	Running	0	3h37m
my-pod2	1/1	Running	0	3h36m
my-rs-ctmbx	0/1	ImagePullBackOff	0	126m
my-rs-fd6z6	0/1	ImagePullBackOff	0	127m
my-rs-h8klx	0/1	ImagePullBackOff	0	127m
my-rs-wcqzw	0/1	ImagePullBackOff	0	127m
mypod	1/1	Running	0	3h40m

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
[sudo] password for arvind_2004:
```

```
arvind_2004@ARVINDM:~/my_project$
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl create deployment webnginx2 --  
image=nginx:latest --replicas=1
```

```
deployment.apps/webnginx2 created
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
my-app1	0/1	ImagePullBackOff	0	145m
my-pod	1/1	Running	0	3h44m
my-pod2	1/1	Running	0	3h43m
my-rs-ctmbx	0/1	ImagePullBackOff	0	133m
my-rs-fd6z6	0/1	ImagePullBackOff	0	134m
my-rs-h8klx	0/1	ImagePullBackOff	0	134m
my-rs-wcqzw	0/1	ImagePullBackOff	0	134m
mypod	1/1	Running	0	3h47m

webnginx2-85b94f8d56-vgh6b 1/1 Running 0 13s

arvind_2004@ARVINDM:~/my_project\$ kubectl get rs

NAME	DESIRED	CURRENT	READY	AGE
------	---------	---------	-------	-----

my-rs	4	4	0	134m
-------	---	---	---	------

webnginx2-85b94f8d56	1	1	1	46s
----------------------	---	---	---	-----

arvind_2004@ARVINDM:~/my_project\$ kubectl kubectl delete all -all

error: unknown command "kubectl" for "kubectl"

arvind_2004@ARVINDM:~/my_project\$ kubect delete all -all

Command 'kubect' not found, did you mean:

command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.3)

Try: sudo apt install <deb name>

arvind_2004@ARVINDM:~/my_project\$ kubectl delete all -all

error: unknown shorthand flag: 'a' in -all

See 'kubectl delete --help' for usage.

arvind_2004@ARVINDM:~/my_project\$ kubectl delete all --all

pod "my-app1" deleted

pod "my-pod" deleted

pod "my-pod2" deleted

pod "my-rs-ctmbx" deleted

pod "my-rs-fd6z6" deleted

pod "my-rs-h8klx" deleted

pod "my-rs-wcqzw" deleted

pod "mypod" deleted

pod "webnginx2-85b94f8d56-vgh6b" deleted

service "kubernetes" deleted

deployment.apps "webnginx2" deleted

replicaset.apps "my-rs" deleted

replicaset.apps "webnginx2-85b94f8d56" deleted

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-rs-kdzk7	0/1	Terminating	0	26s
-------------	-----	-------------	---	-----

my-rs-nnwdc	0/1	Terminating	0	25s
-------------	-----	-------------	---	-----

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

```
\Error from server (BadRequest): error when creating "deployment.yml": Deployment in version "v1" cannot be handled as a Deployment: json: cannot unmarshal string into Go struct field Container.spec.template.spec.containers.ports of type v1.ContainerPort
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

```
Error from server (BadRequest): error when creating "deployment.yml": Deployment in version "v1" cannot be handled as a Deployment: json: cannot unmarshal string into Go struct field Container.spec.template.spec.containers.ports of type v1.ContainerPort
```

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

```
deployment.apps/my-deployment created
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
------	-------	------------	-----------	-----

my-deployment	0/3	3	0	8s
---------------	-----	---	---	----

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-deployment-f47778678-czpqr	0/1	ErrImagePull	0	18s
-------------------------------	-----	--------------	---	-----

my-deployment-f47778678-d5bxg	0/1	ImagePullBackOff	0	18s
-------------------------------	-----	------------------	---	-----

my-deployment-f47778678-d8f7j	0/1	ImagePullBackOff	0	18s
-------------------------------	-----	------------------	---	-----

```
arvind_2004@ARVINDM:~/my_project$ kubectl describe deployment my-deployment
```

```
Name: my-deployment
```

```
Namespace: default
```

```
CreationTimestamp: Thu, 20 Mar 2025 08:37:01 +0000
```

```
Labels: <none>
```

```
Annotations: deployment.kubernetes.io/revision: 1
```

Selector: app=my-app

Replicas: 3 desired | 3 updated | 3 total | 0 available | 3 unavailable

StrategyType: RollingUpdate

MinReadySeconds: 0

RollingUpdateStrategy: 25% max unavailable, 25% max surge

Pod Template:

Labels: app=my-app

Containers:

my-app:

Image: arvindm2004/webapp1:tagname

Port: 8089/TCP

Host Port: 0/TCP

Environment: <none>

Mounts: <none>

Volumes: <none>

Node-Selectors: <none>

Tolerations: <none>

Conditions:

Type	Status	Reason
------	--------	--------

----	-----	-----
------	-------	-------

Available	False	MinimumReplicasUnavailable
-----------	-------	----------------------------

Progressing	True	ReplicaSetUpdated
-------------	------	-------------------

OldReplicaSets: <none>

NewReplicaSet: my-deployment-f47778678 (3/3 replicas created)

Events:

Type	Reason	Age	From	Message
------	--------	-----	------	---------

----	-----	-----	-----	-----
------	-------	-------	-------	-------

Normal ScalingReplicaSet 26s deployment-controller Scaled up replica set my-deployment-f47778678 from 0 to 3

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl create deployment webnginx2 --image=nginx:latest --replicas=1
```

deployment.apps/webnginx2 created

```
arvind_2004@ARVINDM:~/my_project$ kubectl replace -f deployment.yml
```

deployment.apps/my-deployment replaced

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
my-deployment-f47778678-5qspr	0/1	ContainerCreating	0	12s
my-deployment-f47778678-c8d5m	0/1	ErrImagePull	0	12s
my-deployment-f47778678-czpqr	0/1	ImagePullBackOff	0	4m32s
my-deployment-f47778678-d5bxg	0/1	ImagePullBackOff	0	4m32s
my-deployment-f47778678-d8f7j	0/1	ImagePullBackOff	0	4m32s
my-deployment-f47778678-f82gp	0/1	ContainerCreating	0	12s
my-deployment-f47778678-fml6f	0/1	ErrImagePull	0	12s
my-deployment-f47778678-zrh84	0/1	ContainerCreating	0	12s
webnginx2-85b94f8d56-gsz4q	1/1	Running	0	72s

```
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deploy --replicas=2
```

error: no objects passed to scale deployments.apps "my-deploy" not found

```
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy mydeploy --replicas=2
```

error: no objects passed to scale deployments.apps "mydeploy" not found

```
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deploy --replicas=2
```

error: no objects passed to scale deployments.apps "my-deploy" not found

```
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deployment --replicas=2
```

deployment.apps/my-deployment scaled

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

```

my-deployment-f47778678-czpqr 0/1 ImagePullBackOff 0 10m
my-deployment-f47778678-d8f7j 0/1 ImagePullBackOff 0 10m
webnginx2-85b94f8d56-gsz4q 1/1 Running 0 7m13s
arvind_2004@ARVINDM:~/my_project$ kubectl delete webnginx2-85b94f8d56-gsz4q
error: the server doesn't have a resource type "webnginx2-85b94f8d56-gsz4q"
arvind_2004@ARVINDM:~/my_project$ kubectl delete pod webnginx2-85b94f8d56-gsz4q
pod "webnginx2-85b94f8d56-gsz4q" deleted
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME READY STATUS RESTARTS AGE
my-deployment-f47778678-czpqr 0/1 ImagePullBackOff 0 11m
my-deployment-f47778678-d8f7j 0/1 ImagePullBackOff 0 11m
webnginx2-85b94f8d56-cvxx2 1/1 Running 0 6s
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME READY STATUS RESTARTS AGE
my-deployment-f47778678-czpqr 0/1 ImagePullBackOff 0 11m
my-deployment-f47778678-d8f7j 0/1 ErrImagePull 0 11m
webnginx2-85b94f8d56-cvxx2 1/1 Running 0 10s
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME READY STATUS RESTARTS AGE
my-deployment-f47778678-czpqr 0/1 ImagePullBackOff 0 12m
my-deployment-f47778678-d8f7j 0/1 ImagePullBackOff 0 12m
webnginx2-85b94f8d56-cvxx2 1/1 Running 0 29s
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
arvind_2004@ARVINDM:~/my_project$ kubectl create apply -f deployment.yml
error: Unexpected args: [apply]
See 'kubectl create -h' for help and examples
arvind_2004@ARVINDM:~/my_project$ kubectl create apply -f deployment.yml
error: Unexpected args: [apply]

```

See 'kubectl create -h' for help and examples

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

```
deployment.apps/my-deploy created
```

```
Error from server (BadRequest): error when creating "deployment.yml": Service in version "v1" cannot be handled as a Service: json: cannot unmarshal string into Go struct field ServiceSpec.spec.selector of type map[string]string
```

```
arvind_2004@ARVINDM:~/my_project$ cat deployment.yml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: my-deploy
```

```
  labels:
```

```
    name: my-deploy
```

```
spec:
```

```
  replicas: 1
```

```
  selector:
```

```
    matchLabels:
```

```
      apptype: web-backend
```

```
  strategy:
```

```
    type: RollingUpdate
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        apptype: web-backend
```

```
    spec:
```

```
      containers:
```

```
        - name: my-app
```

```
          image: arvindm2004/webapp1:tagname
```

```
      ports:
```

- containerPort: 7070

apiVersion: v1

kind: Service

metadata:

name: my-service

labels:

app: my-service

type: backend-app

spec:

type: NodePort

ports:

- targetPort: 7070

port: 7070

nodePort: 30002

selector:

apptype: web-backend

arvind_2004@ARVINDM:~/my_project\$ sudo nano deployment.yml

[sudo] password for arvind_2004:

arvind_2004@ARVINDM:~/my_project\$ kubectl apply -f deployment.yml

service/my-service created

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
my-deploy-5df65497db-xfhlq	0/1	ErrImagePull	0	3m40s
my-deployment-f47778678-czpqr	0/1	ImagePullBackOff	0	41m
my-deployment-f47778678-d8f7j	0/1	ImagePullBackOff	0	41m
webnginx2-85b94f8d56-cvxx2	1/1	Running	0	29m

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	44m
my-service	NodePort	10.106.84.41	<none>	7070:30002/TCP	31s

```
[sudo] password for arvind_2004:
```

deployment.apps/my-deploy configured

```
arvind_2004@ARVINDM:~/my_project$ kubectl replace deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl replace -f deployment.yml
```

service/my-service replaced

NAMESPACE	NAME	TARGET PORT	URL
default	my-service	7070	http://192.168.49.2:30002

✖ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found

🐾 If the above advice does not help, please let us know:


```
arvind_2004@ARVINDM:~/my_project$ minikube service myservice.yml
```

You may select another namespace by using 'minikube service myservice.yml -n <namespace>'. Or list out all the services using 'minikube service list'

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL
-----	-----	-----	-----
default	my-service	9000	http://192.168.49.2:30002
-----	-----	-----	-----

✖ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found

|  If the above advice does not help, please let us know: |

|  <https://github.com/kubernetes/minikube/issues/new/choose> |

| |

| Please run `minikube logs --file=logs.txt` and attach logs.txt to the GitHub issue. |

| Please also attach the following file to the GitHub issue: |

| - /tmp/minikube_service_8aaa71feca0d33a15dc0b5f1d3de1c519de02edd_0.log |

| |

```
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30002
```

```
curl: (7) Failed to connect to 192.168.49.2 port 30002 after 0 ms: Couldn't connect to server
```

```
arvind_2004@ARVINDM:~/my_project$ cat deployment.yml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: my-deploy
```

```
  labels:
```

```
    name: my-deploy
```

```
spec:
```

```
  replicas: 4
```

```
  selector:
```

```
    matchLabels:
```

```
      apptype: web-backend
```

```
  strategy:
```

```
    type: RollingUpdate
```

```
  template:
```

```
metadata:
  labels:
    apptype: web-backend
spec:
  containers:
  - name: my-app
    image: arvindm2004/webapp1:tagname
    ports:
      - containerPort: 9000
```

```
apiVersion: v1
kind: Service
metadata:
  name: my-service
  labels:
    app: my-service
    type: backend-app
```

```
spec:
  type: NodePort
  ports:
  - targetPort: 9000
    port: 9000
    nodePort: 30002
```

```
selector:
  apptype: web-backend
```

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

```
my-deploy-84f4966d79-64z7p 0/1 ImagePullBackOff 0 4m41s
my-deploy-84f4966d79-7f92t 0/1 ImagePullBackOff 0 4m41s
my-deploy-84f4966d79-bt29f 0/1 ImagePullBackOff 0 4m41s
my-deploy-84f4966d79-n4ts5 0/1 ImagePullBackOff 0 4m41s
```

arvind_2004@ARVINDM:~/my_project\$ kubectl describe deployment my-deploy

Name: my-deploy

Namespace: default

CreationTimestamp: Thu, 20 Mar 2025 10:07:31 +0000

Labels: name=my-deploy

Annotations: deployment.kubernetes.io/revision: 1

Selector: apptype=web-backend

Replicas: 4 desired | 4 updated | 4 total | 0 available | 4 unavailable

StrategyType: RollingUpdate

MinReadySeconds: 0

RollingUpdateStrategy: 25% max unavailable, 25% max surge

Pod Template:

Labels: apptype=web-backend

Containers:

my-app:

Image: arvindm2004/webapp1:tagname

Port: 9000/TCP

Host Port: 0/TCP

Environment: <none>

Mounts: <none>

Volumes: <none>

Node-Selectors: <none>

Tolerations: <none>

Conditions:

Type	Status	Reason
------	--------	--------

----	-----	-----
------	-------	-------

Available	False	MinimumReplicasUnavailable
-----------	-------	----------------------------

Progressing	True	ReplicaSetUpdated
-------------	------	-------------------

OldReplicaSets: <none>

NewReplicaSet: my-deploy-84f4966d79 (4/4 replicas created)

Events:

Type	Reason	Age	From	Message
------	--------	-----	------	---------

----	-----	----	----	-----
------	-------	------	------	-------

Normal	ScalingReplicaSet	4m54s	deployment-controller	Scaled up replica set my-deploy-84f4966d79 from 0 to 4
--------	-------------------	-------	-----------------------	--

arvind_2004@ARVINDM:~/my_project\$ kubectl port-forward service/my-service 9000:9000

error: unable to forward port because pod is not running. Current status=Pending

arvind_2004@ARVINDM:~/my_project\$ kubectl delete all --all

pod "my-deploy-84f4966d79-64z7p" deleted

pod "my-deploy-84f4966d79-7f92t" deleted

pod "my-deploy-84f4966d79-bt29f" deleted

pod "my-deploy-84f4966d79-n4ts5" deleted

service "kubernetes" deleted

service "my-service" deleted

deployment.apps "my-deploy" deleted

arvind_2004@ARVINDM:~/my_project\$ kubectl delete pods --all

pod "my-deploy-84f4966d79-52xlz" deleted

pod "my-deploy-84f4966d79-rptt5" deleted

arvind_2004@ARVINDM:~/my_project\$ kubectl apply -f deployment.yml

deployment.apps/my-deploy created

service/my-service created

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
my-deploy-84f4966d79-qhtb2	0/1	ContainerCreating	0	9s
my-deploy-84f4966d79-tg7nl	0/1	ContainerCreating	0	9s
my-deploy-84f4966d79-zflkr	0/1	ErrImagePull	0	9s
my-deploy-84f4966d79-zgf99	0/1	ContainerCreating	0	9s

```
arvind_2004@ARVINDM:~/my_project$ kubectl port-forward service/my-service 9000:9000
```

error: unable to forward port because pod is not running. Current status=Pending

```
arvind_2004@ARVINDM:~/my_project$ docker pull arvindm2004/webapp1:tagname
```

Error response from daemon: manifest for arvindm2004/webapp1:tagname not found:
manifest unknown: manifest unknown

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

deployment.apps/my-deploy configured

error: error parsing deployment.yml: error converting YAML to JSON: yaml: line 15: mapping values are not allowed in this context

```
arvind_2004@ARVINDM:~/my_project$ minikube service my-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	my-service	9000	http://192.168.49.2:30002

✗ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found

--	--

 If the above advice does not help, please let us know:	
--	--

apptype: web-backend

strategy:

type: RollingUpdate

template:

metadata:

labels:

apptype: web-backend

spec:

containers:

- name: my-app

image: arvindm2004/webapp1:tagname

ports:

- containerPort: 9000

apiVersion: v1

kind: Service

metadata:

name: my-service

labels:

app: my-service

spec:

type: NodePort

ports:

- port: 9000

targetPort: 8080

nodePort: 30002

selector:

apptype: web-backend
apiVersion: apps/v1

kind: Deployment

metadata:

name: my-deploy

labels:

name: my-deploy

spec:

replicas: 1

selector:

matchLabels:

apptype: web-backend

strategy:

type: RollingUpdate

template:

metadata:

labels:

apptype: web-backend

spec:

containers:

- name: my-app

image: arvindm2004/webapp1:tagname

ports:

- containerPort: 9000

apiVersion: v1

kind: Service

metadata:

name: my-service

labels:

app: my-service

spec:

type: NodePort

ports:

- port: 9000

targetPort: 8080

nodePort: 30002

selector:

apptype: web-backend

arvind_2004@ARVINDM:~/my_project\$ sudo nano deployment.yml

arvind_2004@ARVINDM:~/my_project\$ kubectl apply -f deployment.yml

Warning: resource deployments/my-deploy is missing the kubectl.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.

deployment.apps/my-deploy configured

service/my-service configured

arvind_2004@ARVINDM:~/my_project\$ minikube service my-service

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL
-----	-----	-----	-----
default	my-service	9000	http://192.168.49.2:30002
-----	-----	-----	-----

NAMESPACE	NAME	TARGET PORT	URL
default	my-service	9000	http://192.168.49.2:30002

✗ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found

 If the above advice does not help, please let us know:

 <https://github.com/kubernetes/minikube/issues/new/choose>

Please run `minikube logs --file=logs.txt` and attach logs.txt to the GitHub issue.

Please also attach the following file to the GitHub issue:

- /tmp/minikube_service_8aaa71feca0d33a15dc0b5f1d3de1c519de02edd_0.log

```

arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
arvind_2004@ARVINDM:~/my_project$ kubectl delete all --all
pod "my-deploy-84f4966d79-qhtb2" deleted
service "kubernetes" deleted
service "my-service" deleted
deployment.apps "my-deploy" deleted
  
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

```
deployment.apps/my-deploy created
```

```
service/my-service created
```

```
arvind_2004@ARVINDM:~/my_project$ minikube service my-service
```

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL
-----	-----	-----	-----
default	my-service	9001	http://192.168.49.2:30002
-----	-----	-----	-----

✗ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found

```
|
|
| 🐱 If the above advice does not help, please let us know:
| 📖 https://github.com/kubernetes/minikube/issues/new/choose
|
| Please run `minikube logs --file=logs.txt` and attach logs.txt to the GitHub issue.
| Please also attach the following file to the GitHub issue:
| - /tmp/minikube_service_8aaa71feca0d33a15dc0b5f1d3de1c519de02edd_0.log
|
|
|
```

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl delete all --all
```

```
pod "my-deploy-6b478cb7cb-4fhvv" deleted
```

```
service "kubernetes" deleted
```

```
service "my-service" deleted
```

```
deployment.apps "my-deploy" deleted
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

```
deployment.apps/my-deploy created
```

```
service/my-service created
```

```
arvind_2004@ARVINDM:~/my_project$ minikube service my-service
```

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL
-----	-----	-----	-----
default	my-service	9001	http://192.168.49.2:30005
-----	-----	-----	-----

 Starting tunnel for service my-service.

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL
-----	-----	-----	-----
default	my-service		http://127.0.0.1:36227
-----	-----	-----	-----

 Opening service default/my-service in default browser...

 <http://127.0.0.1:36227>

 Because you are using a Docker driver on linux, the terminal needs to be open to run it.

 ^C Stopping tunnel for service my-service.

```
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-
color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not
```

Found</h1><hr class="line" /><p>Type Status Report</p><p>Description
The origin server did not find a current representation for the target resource or is not
willing to disclose that one exists.</p><hr class="line" /><h3>Apache
Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:~/my_project\$ curl
http://192.168.49.2:30005/my-app/

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style  
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b  
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-  
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-  
color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not  
Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b>  
The origin server did not find a current representation for the target resource or is not  
willing to disclose that one exists.</p><hr class="line" /><h3>Apache  
Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:kubect get pod
```

Command 'kubect' not found, did you mean:

command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.3)

Try: sudo apt install <deb name>

arvind_2004@ARVINDM:~/my_project\$ kubectl get pod

NAME	READY	STATUS	RESTARTS	AGE
my-deploy-6b478cb7cb-t5lkg	1/1	Running	0	3m38s

arvind_2004@ARVINDM:~/my_project\$ kubectl exec -it my-deploy-my-deploy-6b478cb7cb-
t5lkg -- /bin/bash

Error from server (NotFound): pods "my-deploy-my-deploy-6b478cb7cb-t5lkg" not found

arvind_2004@ARVINDM:~/my_project\$ minikube service my-service

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL
-----	-----	-----	-----
default	my-service	9001	http://192.168.49.2:30005
-----	-----	-----	-----

 Starting tunnel for service my-service.

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL


```
|-----|-----|-----|-----|
| default | my-service |      | http://127.0.0.1:35123 |
|-----|-----|-----|-----|
```

🐘 Opening service default/my-service in default browser...

👉 http://127.0.0.1:35123

🔔 Because you are using a Docker driver on linux, the terminal needs to be open to run it.

^C👉 Stopping tunnel for service my-service.

```
arvind_2004@ARVINDM:~/my_project$ curl http://127.0.0.1:35123
```

```
curl: (7) Failed to connect to 127.0.0.1 port 35123 after 0 ms: Couldn't connect to server
```

```
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-
color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not
Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b>
The origin server did not find a current representation for the target resource or is not
willing to disclose that one exists.</p><hr class="line" /><h3>Apache
Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:curl
http://192.168.49.2:30005/my-app/
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-
color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not
Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b>
The origin server did not find a current representation for the target resource or is not
willing to disclose that one exists.</p><hr class="line" /><h3>Apache
Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:curl
http://192.168.49.2:30005/my-service/
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-
color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not
```

Found

Type Status Report

Description

The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.

Apache Tomcat/9.0.102

arvind_2004@ARVINDM:~/my_project\$ curl http://192.168.49.2:30005/my-app

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b> The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3></body></html>
```

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
my-deploy-6b478cb7cb-t5lkg	1/1	Running	0	6m50s

arvind_2004@ARVINDM:~/my_project\$ curl http://192.168.49.2:30005/my-deploy

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b> The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3></body></html>
```

arvind_2004@ARVINDM:~\$ curl http://192.168.49.2:30005/my-deploy/

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b> The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3></body></html>
```

arvind_2004@ARVINDM:~/my_project\$ curl http://192.168.49.2:30005/my-deploy/

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-
color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not
Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b>
The origin server did not find a current representation for the target resource or is not
willing to disclose that one exists.</p><hr class="line" /><h3>Apache
Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:curl
http://192.168.49.2:30005/maven-web-app/
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-
color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not
Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b>
The origin server did not find a current representation for the target resource or is not
willing to disclose that one exists.</p><hr class="line" /><h3>Apache
Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:~/my_project$ kubectl exec -
it my-deploy-6b478cb7cb-t5lkg -- /bin/bash
```

```
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat# ls
```

```
bin      conf      filtered-KEYS  LICENSE  native-jni-lib  README.md  RUNNING.txt
upstream-KEYS  webapps.dist
```

```
BUILDING.txt  CONTRIBUTING.md  lib      logs  NOTICE  RELEASE-NOTES  temp
webapps      work
```

```
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat# cd webapps
```

```
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat/webapps# ls
```

```
my-web  my-web.war
```

```
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat/webapps# cd
```

```
root@my-deploy-6b478cb7cb-t5lkg:~# exit
```

```
exit
```

```
arvind_2004@ARVINDM:~/my_project$ ls
```

```
deployment.yml  kubectl.txt  my-pod.yml  pod2.yml  rs-test.yml
```

```
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-web/
```

```
<html>
```

```
<body>
```

```
<h2>Hello World!</h2>
```

```
</body>
```

```
</html>
```

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-web/
```

```
<html>
```

```
<body>
```

```
<h2>Hello World!</h2>
```

```
</body>
```

```
</html>
```